

RESEARCH INFORMED CONSENT FORM

STUDY TITLE: EEG/MEG Studies of Human Memory (IRB-FY2024-8635)

INVESTIGATOR(S): Sebastian Michelmann

SUMMARY OF THE RESEARCH

You have been invited to take part in a study to learn more about **how the brain processes, stores, and retrieves memories over time**. If you agree to be in this study, you will be asked to do the following:

- Lie on a table that will be slid into the MRI scanner so that your head and upper body are inside the magnet tube.
- Wear headphones to limit the noise from the MRI and allow you to hear audio and instructions for the tasks you are being asked to do, or questions from the investigator or the MRI operator.
- Have foam pads placed around your head to help you hold your head still during the MRI scan, so that the scans will be clear.
- Lie still throughout your time in the MRI scanner (up to 1 hour).

Participation in this study will involve one MRI session (1 hour or less) and no more than (1 hour) total. There are no known risks associated with your participation in this research beyond those of everyday life. You are not expected to benefit from participating in this research.

If you have a fear of enclosed spaces (*claustrophobia*), you likely will **not** want to participate in the research.

Participation in this study is voluntary. You may refuse to participate or withdraw at any time without penalty. Please read the rest of this consent form for more information about the study.

INVITATION TO PARTICIPATE IN A RESEARCH STUDY

You are invited to participate in a research study. This form has information to help you decide whether or not you wish to participate—please review it carefully. Your participation is voluntary. Please ask any questions you have about the study or about this form before deciding to participate.

PURPOSE OF THE STUDY

The purpose of this study is to find out how the everyday knowledge we already possess (e.g., the typical sequence of events in a restaurant: arrive → order → eat → pay → leave) interacts with newly learned experiences of similar situations (e.g., a new restaurant visit) to guide our memory processes. By understanding these interactions, we aim to learn more about how memories are stored and retrieved.

PROCEDURES

If you agree to be in this study, images of your brain, called scans, will be taken using a Magnetic Resonance Imaging (MRI) scanner. The MRI scanner is a machine that enables us to acquire images of the brain using the changing magnetic fields inside your body.

If you agree to take part in the study, you will be asked to do the following:

- Lie on a table that will be slid into the MRI scanner (so that your head and upper body are inside the magnet tube);
- Wear headphones to limit the noise from the MRI and allow you to hear audio and instructions for the tasks you are being asked to do, or questions from the investigator or the MRI operator.
- Have foam pads placed around your head to help you hold your head still during the MRI scan, so that the scans will be clear.
- Lie still throughout your time in the MRI scanner.
- Allow a special video camera to record your eye movements and blinking during the scan.
- Complete a brief calibration procedure: you will look at a series of dots on the screen so the system can accurately track your gaze coordinates on the screen.
- Complete the main experimental task (see below)
- Complete the post-experiment survey.

Anatomical images (images that show us the structures in your brain) will be obtained during the first **5** minutes of the scan. After the anatomical images, functional images will be obtained **for approximately 45 minutes**. Functional images are scans that show us how the brain works by illustrating what the brain is actively doing at a particular time. During this time, you will be asked to:

- View visual stimuli, listen to auditory stimuli (e.g., stories, words), or learn associations between stimuli (e.g., words, images, videos, audios, stories, or symbols)
- Perform simple tasks (e.g., counting down in steps of 13, doing simple math, identifying atypical stimuli in a series of typical stimuli, or making judgments on odd/even numbers)
- Identify moments within a story/video, or recall the learned stimuli (e.g., freely reporting what you learned, identifying associates from a series of options, deciding whether a stimulus has been previously seen, judging the similarity of a stimulus to previously seen ones, or deciding whether a shown association is intact or has been rearranged)

Participation in this study will involve **1** hour of your time with **0.85** hours for the MRI scans and 0.15 hours for study set-up.

During your MRI scanning session, occasional breaks of a few minutes will be provided. You will remain in the MRI scanner during breaks, but you may close your eyes and rest. You may rest for as long as you need to be ready to continue the session.

ELIGIBILITY

You must be 18 years or older to participate in this study. If you are under the care of an epileptologist, you must have doctors' consent in writing in order to participate.

You should not participate in this study if you have any of the following:

1. Surgical aneurysm clips
2. Cardiac pacemaker (even if removed)
3. Prosthetic heart valve
4. Neurostimulator
5. Implanted pumps
6. Cochlear implants
7. History of injury to the eye involving metal

Each of these is related to potential harm during the MRI scan.

Additionally, if you have any of the following, please notify the researcher:

1. History of head trauma
2. Transdermal patch (nicotine, nitroglycerin, hormone)
3. Metal rod, plates, or screws
4. Previous surgeries
5. IUD (Intrauterine device)
6. Hearing aid
7. Dentures, braces, or a non-removable retainer
8. Hair Extensions or wig/hairpiece
9. Any metal in the body (shrapnel, bullets, or buckshot)
10. Colored contact lenses or magnetic eyelashes
11. Facial tattoos or tattoos that cover more than 5% of your body or are larger than 8 inches
12. Worked as a welder or metal worker

This does not necessarily exclude you from the study, but should be disclosed in advance, as they may cause you mild discomfort or affect the MRI images.

Additionally, you **may** participate in this study if you:

1. are proficient in English: able to read and answer questions in English, and able to understand and follow instructions presented in English
2. have normal or corrected-to-normal vision (non-colored contact lenses are accepted, or we will provide MRI-safe glasses)
3. are willing to have your eye movements recorded using an eye-tracking device.

Additionally, you **may not** participate in this study if you:

4. are not proficient in English: not comfortable reading and answering questions in English, or unable to understand and follow instructions presented in English
5. have a visual or eye condition that may interfere with accurate eye tracking (e.g., uncontrolled eye movements, severe visual impairment)

RISKS OR DISCOMFORTS

There are no known significant risks or side effects associated with MRI scans. The magnetic fields at the used strengths fall within the Food and Drug Administration [FDA] guidelines for radiofrequency electromagnetic field exposure from MRI systems.

There is a risk if metal objects are near the MRI because they can be drawn into the MRI scanner, and that could hurt someone in or near the machine. Metal objects might be in a person's body if they have electrically, magnetically, or mechanically activated implants (such as cardiac pacemakers), clips on blood vessels in their brain, or metallic fragments in their eye. To protect against this risk, you will be carefully screened for previous exposure to metallic fragments or to implanted devices. You will also be asked to place all metallic and magnetic objects in your possession (e.g., keys, jewelry, credit cards) in a locker outside the MRI room.

The Center for Brain Imaging uses software that is not regulated by the FDA because it is intended for research purposes and not for diagnostic purposes.

Although there are no known risks of an MRI scan to the unborn fetus, we will not let you take part in the study if you are or might be pregnant. We will have you take a urine pregnancy test right before the scan to find out if you are pregnant. If the test shows that you are pregnant, we will cancel the scan. Results will only be reported to you, as this is considered private health information.

If you would like to decline this pregnancy test, please initial here:

____ My initials indicate that I *choose not* to take a urine pregnancy test and will still be scanned.

Some people have reported mild discomfort during MRI scans, such as:

- Claustrophobia (fear of enclosed spaces). You will be asked to lie on a table that slides into a horizontal cylinder only slightly wider than your body in all directions, and your head will be secured to help you stay still. If you are likely to be uncomfortable or afraid in enclosed spaces, you will not be allowed to participate.
- Mild back discomfort. If you have ever had back surgery or have a history of chronic back pain, please notify the researcher in charge of the scan so the padding on the MRI table can be enhanced.
- Reaction to noise levels. The MRI scanner makes loud knocking or beeping sounds during scans; earplugs and/or headphones will be provided to help reduce this noise.
- Peripheral nerve stimulation. Because the magnet turns on and off during the scan, the possibility exists for peripheral nerve stimulation. If this happens, you may feel creeping or tingling sensations, typically along your arms or lower back.
- Dizziness and nausea may occur while you are in the MRI.
- Drying of eye contact lenses. This rarely occurs, but you should notify the investigator that you have contact lenses in your eyes, and you should either moisten your eyes before entering the scanner or remove your contacts. MRI safe glasses are available for use if needed, with lenses that accommodate the most common prescriptions.
- Feeling some warmth from the radio frequency coils can occur. The MRI scanner is set so that this warmth will not raise the body's temperature by more than 1 degree above normal.

You may notify the research staff at any time if you feel uncomfortable, no matter what the reason. Between each scan, you will be in contact with the research staff through a microphone mounted in the MRI scanner. However, while the scanner is on, the speaker will be turned off, and the research staff will not be able to hear you. Nonetheless, in the case of an emergency, you will have a handheld squeeze ball device to let the operator know if you wish to immediately stop scanning and be removed from the magnet. The MRI scan can be stopped at any time at your request; when you squeeze the squeeze ball, the research staff will be notified and stop the scan. If you think that you have experienced a research-related injury call **Sebastian Michelmann, (609) 285-6734, s.michelmann@nyu.edu**. Please be aware that emergency medical services are not available on-site.

Although unlikely, there is a small chance that private information about you that is collected in this research may be disclosed to individuals that you would not want to have access to (i.e., a breach of confidentiality).

New York University will not provide either medical treatment or financial compensation for any injury resulting from your participation in this research. You do not give up any legal rights to seek payment for injury by consenting to this research.

BENEFITS

Although you will receive no direct benefits, this research may help contribute to knowledge about how our “schemas” (everyday, familiar knowledge) influence what we remember and what we forget.

INCIDENTAL FINDINGS

The investigators for this project are not trained to perform radiological diagnosis, and the MRI scans performed in this study are not designed to find abnormalities. The investigators and NYU are not responsible for failure to find existing abnormalities in

your MRI scans. However, on occasion, the investigator may notice a finding on an MRI scan that seems abnormal. If you agree below, and if this occurs within 12 months of the scan, a neuroradiologist will be consulted, and the MRI will be sent to NYU Medical Center. This MRI scan, along with the consulting neuroradiologist's report, will be part of your permanent medical record. If the neuroradiologist thinks that further investigation of the finding is called for, the investigator will contact you and inform you of the finding. The decision as to whether to proceed with further examination or treatment lies solely with you and your physician. The investigators, the consulting neuroradiologist, and NYU are not responsible for any examination or treatment that you undertake based upon these findings. Because the images collected in this study do not constitute a proper clinical MRI study, these images will not be made available for diagnostic purposes.

Please initial below to indicate your decision:

☐ Yes, I consent to my MRI scan being shared with a neuroradiologist if the investigator notices a finding on the MRI scan that seems abnormal. I understand that the MRI scan, along with the consulting neuroradiologist's report, will be part of my permanent medical record.

☐ No, I do not consent to my MRI scan being shared with a neuroradiologist and I understand that I will *not* be notified if the investigator notices a finding on the MRI scan that seems abnormal.

ACCESS TO YOUR STUDY INFORMATION

Other than the incidental findings described above, you **will** have access to the anatomical MRI scan that is collected from you in this study. By requesting directly from the study team, participants will be sent a copy of their anatomical MRI via email. Participants will be made aware that the images generated during the scan should not be used as diagnostic criteria, as they do not comprise that of a proper clinical MRI.

COST AND COMPENSATION

You will be paid **\$30/hour** for completing the MRI scanning session; should you withdraw before the end of the study, partial payment for the time you have completed will be given. There will be no cost to you associated with participation in this study.

If you are paid more than \$2000 from New York University within any calendar year, NYU will be required to report the payments to the IRS and will issue you a 1099 form, which means that you may be expected to pay taxes on the payments.

VOLUNTARY PARTICIPATION

Participation in this study is voluntary. You may refuse to participate or withdraw at any time without penalty.

We may end your participation in the study if there is loss of compliance with the study protocol or if there are changes in inclusion/exclusion criteria.

PRIVACY & DATA CONFIDENTIALITY

In this study, you may be asked to provide information that could be used to identify you personally. This information will be kept confidential. Only researchers and others who will keep the information confidential (e.g., regulatory agencies or oversight groups) may access information that could personally identify you. Confidentiality of your research records will be strictly maintained. Information will be stored in the investigator's file and will be identified by a code number. The code key connecting your name to specific information about you will be kept in a separate, secure location. Information not containing identifiers may be used in future research or shared with other researchers without your additional consent.

The results of this study may be published in a book, journal, or used for teaching purposes. However, information that could identify you will never be shared unless you explicitly provide written authorization, which will be requested in writing. The MRI images of your brain may be shared anonymously (that is, without any way of identifying who the scan is of) for teaching purposes, research (such as in academic databases), or displayed at scientific conferences or in publications.

Future Use of Data

Information about you collected for this study may be shared with other researchers, used for other research studies, or placed in a data repository. These studies may be similar to this study or completely different. All information that could identify you will be removed before sharing the data or using it for other research studies.

FUTURE RESEARCH

I authorize the principal investigator, **Sebastian Michelmann**, and their co-investigators to contact me about future research on brain function within the NYU Center for Brain Imaging. If I agree, a researcher affiliated with the NYU Center for Brain Imaging might contact me in the future to tell me about a research study. At that time, I can decide whether I am interested in participating in a particular study or not. I will then have the opportunity to contact the researcher to schedule an appointment to be fully informed about the research project.

☐ I agree to be contacted by a researcher affiliated with the NYU Center for Brain Imaging about research studies conducted at the NYU Center for Brain Imaging.

OR

☐ I do not want to be contacted by a researcher affiliated with the NYU Center for Brain Imaging.

CONTACT INFORMATION

You are encouraged to ask questions at any time during this study. For information about the study, contact **Sebastian Michelmann, s.michelmann@nyu.edu, (609) 285-6734**.

If you have questions about your rights as a research participant or if you believe you have been harmed by the research, please contact the NYU Human Research Protection Program at (212) 998-4808 or ask.humansubjects@nyu.edu.

AGREEMENT TO PARTICIPATE

Participant: By signing this document, you are agreeing to participate in this study. Make sure you understand what the study involves before you sign. If you have any questions about the study after you agree to participate, you can contact the research team using the information provided above. You may keep a copy of this form.

Name of Participant (print) _____

Signature of Participant

Date

Person Obtaining Consent: Your signature below means that you have explained the research to the participant and have answered any questions they may have about the research.

Name of Person Obtaining Consent (print) _____

Signature of Person Obtaining Consent

Date